ADVANCED APTITUDE · SESSION 2 OF 7

Time-work & time-speed-distance

Module: Advanced Quant & Speed · Time & work, pipes & cisterns, trains, boats, relative speed

ONE IDEA FOR TODAY

Every question today runs on one equation: output = rate × time.

Work

Units done = efficiency × days

Pipes

Water in tank = net flow × hours

Travel

Distance = speed × time

Today’s target: any work or TSD question in 90 seconds or less.

ACTIVITY · 10 MIN

Diagnostic speed test

How fast are you today?

10 questions

Write the final answer only.

10 minutes

No calculator. Rough work allowed.

90-second rule

Stuck for 90 seconds? Skip and come back.

Finished early? Raise your hand. First three names go on the board.

DIAGNOSTIC · QUESTIONS 1–5

Time & work, pipes

Q1	A finishes a job in 15 days, B in 10 days. Together?
Q2	6 people finish a job in 10 days. How many days for 15 people?
Q3	A is twice as efficient as B. B alone takes 24 days. Together?
Q4	Pipe A fills a tank in 12 min. Pipe B empties it in 18 min. Both open?
Q5	Two pipes fill a tank in 20 h and 30 h. Together?

DIAGNOSTIC · QUESTIONS 6–10

Speed, trains, boats

Q6	90 km/h = ? m/s
Q7	A 180 m train runs at 54 km/h. Time to cross a pole?
Q8	Trains of 120 m and 180 m run in opposite directions at 40 and 50 km/h. Time to cross each other?
Q9	Boat 12 km/h in still water, stream 4 km/h. Time for 24 km upstream?
Q10	You drive to your Bengaluru office at 30 km/h and back at 60 km/h. Average speed?

CHECK YOUR SCORE

Which band are you in?

9–10 CORRECT

Sprinter

Your job today: cut your time further.

6–8 CORRECT

Steady

Your job today: fix the 2–4 you missed.

0–5 CORRECT

Builder

Your job today: own the eight frameworks.

Hands up for each band. Write your score and finish time in your error log.

FRAMEWORK 1

Think in units, not fractions

THE FRAME

1. Total work = LCM of the days.
2. Each person's rate = LCM ÷ their days.
3. Time together = total ÷ sum of rates.

WORKED EXAMPLE

A can paint a 2BHK flat in 12 days, B in 18 days.

Total = LCM(12, 18) = 36 units

A = 3 units/day · B = 2 units/day

36 ÷ 5 = 7.2 days

Your turn: A finishes a job in 10 days, B in 15 days. How long together?

FRAMEWORK 2

Man-days: balance both sides

THE FRAME

$$M_1 \times D_1 \times H_1 \div W_1 = M_2 \times D_2 \times H_2 \div W_2$$

More people ⟶ fewer days. More work ⟶ more days.

Efficiency and time are inverse: twice as efficient = half the time.

WORKED EXAMPLE

A road crew of 15 lays 600 m of highway in 20 days, 8 h a day. Days for 20 workers, 6 h a day, 900 m?

$15 \times 20 \times 8 \div 600 = 20 \times D \times 6 \div 900$

$4 = 120D \div 900 \longrightarrow D = 30$ days

Your turn: 10 packers at a Flipkart fulfilment centre clear 400 orders in 4 hours. How many packers clear 600 orders in 3 hours?

FRAMEWORK 3

When someone leaves or joins

THE FRAME

1. Convert to units first.
2. Fill the work in stages: together, then whoever stays.
3. Remaining units ÷ the remaining rate.

WORKED EXAMPLE

A takes 20 days, B 30 days. They start together; A leaves after 6 days.

Total 60 units · A = 3/day · B = 2/day

6 days together = 30 units done

B finishes 30 ÷ 2 = 15 days → 21 days total

Your turn: A takes 10 days, B 20 days. They work together for 4 days, then A leaves. How many more days for B?

FRAMEWORK 4

Pipes & cisterns: inlets add, outlets subtract

THE FRAME

1. Tank capacity = LCM of the hours.
2. Inlets are + units. Outlets and leaks are − units.
3. Time = capacity ÷ net rate.

WORKED EXAMPLE

A fills a tank in 10 h, B in 15 h, C empties it in 30 h. All open.

Capacity = 30 units

Net = +3 + 2 − 1 = 4 units/h

30 ÷ 4 = 7.5 h

Your turn: A tap fills a tank in 6 h. Because of a leak it takes 8 h. How long does the leak alone take to empty the full tank?

GAME · 3 MIN

Units race

A: 6 days, B: 12 days. Together?

A: 15, B: 10, C: 30 days. All three?

Pipe fills in 4 h, outlet empties in 6 h. Both open?

A is twice as fast as B. Together 10 days. B alone?

Mentor calls a card. First row to shout the answer and the LCM wins the point.

FRAMEWORK 5

Speed basics and average speed

THE FRAME

km/h ⟶ m/s: × 5/18

m/s ⟶ km/h: × 18/5

Same distance both ways:

Average = 2xy ÷ (x + y)

WORKED EXAMPLE

Mumbai to Pune at 60 km/h, back at 40 km/h. Average speed?

Never average the two speeds: 50 is wrong.

2 × 60 × 40 ÷ (60 + 40)

= 48 km/h

Your turn: You cycle to college at 12 km/h and back at 18 km/h. Average speed?

FRAMEWORK 6

Relative speed: add or subtract

OPPOSITE DIRECTIONS

Relative speed = a + b

Two buses 150 km apart leave Mumbai and Pune towards each other at 40 and 35 km/h.

150 ÷ 75 = 2 h to meet

SAME DIRECTION

Relative speed = a − b

A thief is 2 km ahead at 8 km/h. Police follow at 10 km/h.

2 ÷ 2 = 1 h to catch

Your turn: Rahul leaves at 9:00 at 40 km/h. Priya follows on the same road at 9:30 at 60 km/h. When does she catch him?

FRAMEWORK 7

Trains: what length, which speed

Train crosses...	Distance to cover	Speed to use
A pole or a person	Train length	Train speed
A platform or bridge	Train + platform	Train speed
A train, opposite way	$L_1 + L_2$	Sum of speeds
A train, same way	$L_1 + L_2$	Difference of speeds

Example: a 300 m train at 54 km/h (15 m/s) crosses a 450 m platform at Dadar ⟶ 750 ÷ 15 = **50 s**

Your turn: Trains of 200 m and 250 m run in opposite directions at 50 and 40 km/h. Time to cross each other?

FRAMEWORK 8

Boats & streams

THE FRAME

Downstream D = u + v

Upstream U = u − v

u = boat in still water · v = stream

u = (D + U) ÷ 2 · v = (D − U) ÷ 2

WORKED EXAMPLE

A houseboat on the Kerala backwaters does 15 km/h downstream and 9 km/h upstream.

u = (15 + 9) ÷ 2

Boat = 12 km/h · Stream = 3 km/h

Your turn: A boat covers 24 km downstream in 2 h and the same 24 km upstream in 3 h. Speed of the stream?

GAME · 6 MIN

TSD speed duel

72 km/h = ? m/s	150 m train at 36 km/h passes a pole	Downstream 20, upstream 14. Stream?
Out at 30 km/h, back at 20 km/h. Average?	10 km apart, walking towards each other at 3 and 2 km/h	25 m/s = ? km/h

Two rows face off. First correct answer with the framework named wins the point.

Break

10 minutes. Back at [time] for graded practice.

GRADED PRACTICE · LEVEL 1 · PAIRS · 6 MIN

Moderate: apply the frameworks

Q1	A can do a job in 8 days, B in 12 days. How long together?
Q2	Two pipes fill a tank in 20 and 30 min; an outlet empties it in 15 min. All open, time to fill?
Q3	A 240 m train at 72 km/h crosses a 160 m bridge. Time taken?
Q4	Boat speed 10 km/h in still water, stream 2 km/h. Time for 48 km downstream?

GRADED PRACTICE · LEVEL 2 · SOLO · 8 MIN

Advanced: placement-test difficulty

Q5 A + B finish in 12 days, B + C in 15, A + C in 20. How long for A, B and C together?

Q6 A is 50% more efficient than B. Together they finish in 18 days. A alone?

Q7 A train passes a pole in 15 s and a 100 m platform in 25 s. Length of the train?

Q8 A man rows 48 km and back in 14 h. He rows 4 km downstream in the time he rows 3 km upstream. Stream speed?

SECTIONAL TEST · 25 MIN

Tag first, then solve

PASS 1

15 min

Solve every sure-shot. Skip anything over 90 seconds.

PASS 2

7 min

Return to skipped questions. Easiest first.

FINAL

3 min

Check marked answers. Fill blanks only if there is no negative marking.

20 questions: time & work 6 · pipes 4 · speed & relative speed 4 · trains 3 · boats 3

Before solving, write W, P or T next to each question number.

REVIEW

Update your error log

Q no.	Framework	Why I missed it	My fix
4	Pipes	Concept: took the leak as +	Mark outlets as − units
11	Trains	Speed: forgot km/h to m/s	Convert × 5/18 first
17	Boats	Careless: gave boat speed	Underline what is asked

Tag every miss: Concept, Speed or Careless. Hands up: which tag did you get most?

WRAP-UP

What you take away today

EIGHT FRAMEWORKS

LCM units · Man-days · Leave & join · Net pipe rate · Speed basics · Relative speed · Trains · Boats

BEFORE SESSION 3

30 questions in 30 minutes on the practice platform. Update your error log. Memorise × 5/18 and × 18/5.

Next session: [Session 3 topic]